

Session 3 Quiz -What is Machine Learning?

Track B | Grades 9–12 | 20 Questions

Q1 [Type: MCQ | Difficulty: Easy | QID : 52518]

_____ is a subset of AI where computers use data to learn and improve on their own, rather than being explicitly programmed for every single situation.

- A. Machine Learning
- B. Rule-based Logic
- C. Manual Instruction
- D. Fixed Computation

Correct Answer: A

Explanation: Machine Learning focuses on identifying patterns in data to improve performance without explicit programming for every scenario. Rule-based logic, manual instruction, and fixed computation all describe traditional programming, which is the opposite approach -humans write the rules step by step.

Q2 [Type: MCQ | Difficulty: Easy | QID : 52522]

When a Machine Learning model learns that "mouths curved upward" is a common feature of happy faces across thousands of training examples, this process is called _____.

- A. Rule-based logic
- B. Data collection
- C. Manual instruction
- D. Pattern recognition

Correct Answer: D

Explanation: Identifying recurring features such as "curved upward mouths" across many examples is the definition of pattern recognition. The session's key insight is that this is what ML does at its core, just with much larger and more complex data than the happy-face example. Data collection is the step before pattern recognition; rule-based logic and manual instruction describe traditional programming, not ML.

Q3 [Type: MCQ | Difficulty: Medium | QID : 52524]

In Google Teachable Machine, the _____ is a percentage showing how sure the AI is about its prediction based on the patterns it learned during training.

- A. Training count
- B. Accuracy guarantee
- C. Confidence score
- D. Processing speed

Correct Answer: C

Explanation: The confidence score indicates how closely the new data matches the patterns identified during training. Critically, high confidence is not the same as being correct - a model can be very confident and still wrong.

Q4 [Type: MCQ | Difficulty: Easy | QID : 52526]

The goal of generalization in Machine Learning is for the trained model to recognise patterns in NEW data it has never seen before.

- A. True
- B. False

Correct Answer: A (True)

Explanation: Generalization is the ability of a model to apply learned patterns to unfamiliar examples correctly - which is exactly the point of training in the first place. A model that can only recognise the exact images it was trained on has not actually learned anything useful. The whole point of testing the Teachable Machine model on new gestures at the end of the demo is to check whether it has generalized.

Q5 [Type: MCQ | Difficulty: Medium | QID : 52529]

Any system that makes a "prediction" - such as a weather app that says rain is likely must be using Machine Learning.

A. True

B. False

Correct Answer: B (False)

Explanation: The act of making a prediction does not make a system ML. What matters is whether the conditions were learned from data or written by humans. A weather app that follows the rule "if humidity > 85% AND pressure is dropping, predict rain" is rule-based -humans wrote those thresholds. A weather app that learns from years of historical weather data without being told what counts as rainy is learning-based. The vocabulary of "predicting" gets used for both, which is why the distinction can be confusing.

Q6 [Type: MCQ | Difficulty: Medium | QID :]

When a Machine Learning model correctly recognises a peace sign, it does not "understand" what a peace sign means - it has only learned mathematical patterns in pixel data that correlate with the label.

A. True

B. False

Correct Answer: A (True)

Explanation: ML models identify mathematical relationships in data points; they do not grasp the human concepts, intentions, or social meanings behind those data. The Teachable Machine demo recognises gestures purely from pixel patterns -it has no idea what a peace sign symbolises or why people use it. This is also why the same model fails completely on gestures it was not trained on.

Q7 [Type: MCQ | Difficulty: Easy | QID : 52534]

Which of these is a characteristic of traditional programming?

- A.** The system improves its own performance over time with more examples
- B.** The computer executes human-written instructions exactly
- C.** The computer finds its own patterns in the data provided
- D.** It is best suited for complex tasks like recognising sarcasm

Correct Answer: B

Explanation: Traditional programming relies on a human programmer providing explicit, step-by-step logic. A calculator, traffic light, microwave timer, and alarm clock all work this way -the system does exactly what it was told to do, the same way, every time. The other options describe Machine Learning, not traditional programming.

Q8 [Type: MCQ | Difficulty: Easy | QID : 52537]

What are "patterns" in the context of Machine Learning?

- A.** The physical hardware used to store information
- B.** Regular, repeating relationships found within data
- C.** Random pieces of data that have no relationship to each other
- D.** Instructions that a human writes into a script

Correct Answer: B

Explanation: ML algorithms search for repeating regularities to make sense of complex datasets. Patterns can be obvious (numbers increasing by 2 each time), subtle (Fridays at 5pm tend to be congested), or hidden in millions of data points. Hardware, randomness, and scripted rules are all the opposite of what ML is looking for.

Q9 [Type: MCQ | Difficulty: Medium | QID : 52540]

Why would a rule-based system likely fail at filtering modern spam emails?

- A. Spam filters require math that rule-based systems cannot perform
- B. Rule-based systems are too slow to process large amounts of email
- C. Computers cannot read text without Machine Learning
- D. Spammers keep changing tactics, requiring endless manual rules

Correct Answer: D

Explanation: As spammers invent new tricks - adding spaces between letters in "FREE MONEY," rewording phrases, pretending to be someone you know - humans cannot keep up with the millions of rules that would be needed. The other options either invent fake technical limits or get the cause-and-effect backwards.

Q10 [Type: MCQ | Difficulty: Medium | QID : 52543]

A Teachable Machine model is trained on three gestures: peace sign, fist, and high five. A student then shows it a thumbs-up. What will the model most likely do?

- A. Recognise it correctly as a thumbs-up based on visual similarity to known gestures
- B. Refuse to make any prediction since the gesture was not in its training data
- C. Predict one of its three trained classes, often with low confidence
- D. Crash and restart because it cannot handle unfamiliar input

Correct Answer: C

Explanation: Models can only output classes they were trained on -there is no built-in "none of the above" option. The model is forced to pick the closest match among what it knows, and its confidence often drops to reflect uncertainty. It cannot recognise a thumbs-up (it was never trained on one), it will not refuse to answer, and it will not crash.

Q11 [Type: MCQ | Difficulty: Medium | QID : 52547]

Which of these tasks would be considered nearly "impossible" for a standard rule-based system to handle effectively?

- A. Calculating the square root of a very large number
- B. Ringing an alarm at a specific time set by the user
- C. Changing a traffic light from green to yellow after exactly 30 seconds
- D. Understanding sarcasm in a text message

Correct Answer: D

Explanation: Sarcasm depends on tone, context, and relationships, which cannot be captured by fixed, literal rules. The other three options are exactly the kinds of fixed, predictable computations that rule-based programs handle perfectly.

Q12 [Type: MCQ | Difficulty: Medium | QID : 52549]

In the Machine Learning training process, what is the role of the "Algorithm"?

- A. It is the physical camera or sensor used to capture training data from the environment
- B. It is the confidence score that is shown to the user at the end of the process
- C. It is the learning method that looks for similarities and patterns in the training data
- D. It is the collection of labelled photos and examples used to teach the computer

Correct Answer: C

Explanation: The algorithm is the "recipe" or set of instructions that the computer uses to process data into a model. It is not the data itself.

Q13 [Type: MCQ | Difficulty: Medium | QID : 52555]

How does Google Maps predict your travel time and suggest the fastest routes?

- A.** It learns from historical traffic patterns, real-time GPS data from millions of phones
- B.** It uses a timer that assumes every car travels at the speed limit at all times
- C.** Programmers manually update the traffic status for every street in the world every minute
- D.** It follows a fixed rule where highways are always faster than side streets

Correct Answer: A

Explanation: By combining historical patterns (e.g., "Fridays at 5pm this highway is usually congested") with current GPS data from millions of users, Google Maps makes a learning-based prediction. The other options describe rule-based systems that would either be wildly inaccurate (B, D) or operationally impossible at scale (C).

Q14 [Type: MCQ | Difficulty: Medium | QID : 52560]

At its core, Machine Learning is the practice of:

- A.** Writing more efficient rules for computers to follow
- B.** Storing massive amounts of information for quick retrieval
- C.** Translating between human languages
- D.** Finding repeating patterns in data

Correct Answer: D

Explanation: Everything in ML, no matter how complex the math or impressive the result, reduces to pattern recognition. A describes traditional programming. C describes a database. D describes one specific ML application (translation), not the core idea behind all ML.

Q15 [Type: MCQ | Difficulty: Hard | QID : 52562]

A student loads 500 labelled photos of dogs and 500 of cats into Teachable Machine and clicks "Train Model." The software processes for several seconds. What is most likely happening during those seconds?

- A.** The algorithm is adjusting internal parameters to find patterns that separate dogs from cats
- B.** The model is being uploaded to a remote cloud server for permanent storage
- C.** The labels are being shuffled randomly across the photos
- D.** The model is being tested on the exact same photos used for training

Correct Answer: A

Explanation: Training means tuning internal "knobs" inside the algorithm to capture the differences between classes -this is the guess-check-adjust loop running thousands of times. Uploading (B) is not training. Shuffling labels (C) would destroy the learning signal. Testing on training data (D) would happen after training, and is not the same activity.

Q16 [Type: MCQ | Difficulty: Hard | QID : 52564]

Which option best explains why a chess-playing Machine Learning model that beats grandmasters still does not "understand" chess?

- A.** Only software, not hardware, is capable of understanding concepts
- B.** It has memorised every possible chess game from a large database
- C.** It detects patterns that correlate with winning - has no concept of what winning means
- D.** ML models cannot understand chess until they also learn checkers and other board games

Correct Answer: C

Explanation: This is "pattern recognition vs. true understanding" applied at a superhuman scale. Even when an ML model outperforms every human in a domain, it has not developed comprehension -it has found mathematical patterns that statistically

lead to winning positions. A is a fake distinction. B is mathematically impossible (chess has more positions than atoms in the universe). D is an invented constraint.

Q17 [Type: MCQ Scenario | Difficulty: Medium | QID : 52566]

Scenario: A student trains Teachable Machine to recognise her cat using 30 photos -all taken in her bedroom during daytime. When she later tries the model in her kitchen at night, it fails to recognise the same cat.

Question: What is the most likely reason?

- A. The cat looks biologically different in different rooms
- B. The training data lacked variety in lighting and background
- C. Cats become unrecognisable to all ML models in low light
- D. The kitchen's electronics interfere with the phone's camera

Correct Answer: B

Explanation: This mirrors the session's repeated warning: variety in training data matters. The model picked up on whatever signals reliably separated "cat" from "not cat" -and in this case, bedroom walls and daytime lighting were always present, so the model learned them as part of the pattern. A is biologically nonsensical. C and D invent fake technical excuses for what is really a data-variety problem.

Q18 [Type: MCQ Scenario | Difficulty: Hard | QID : 52568]

Scenario: A class trains Teachable Machine on "thumbs up" and "thumbs down." During training, every thumbs-up photo includes a student's red shirt sleeve in the frame, and every thumbs-down photo includes a blue sleeve. The model scores perfectly on the training set but fails completely on new photos with different clothing.

Question: What is the most precise diagnosis?

- A.** The model is broken and needs to be retrained on a faster computer
- B.** The model learned to associate sleeve colour with the labels -a pattern that does not generalise
- C.** The students need to record more photos under exactly the same conditions
- D.** Thumbs-up and thumbs-down gestures simply cannot be learned by ML

Correct Answer: B

Explanation: The model picked up the easiest pattern that separated the two classes - sleeve colour -instead of the hand position. This is the data-variety failure from the session, in its sharpest form: the AI does not "know" which feature is supposed to matter, so it locks onto whatever correlates most reliably with the label. A blames hardware. C would make the problem worse, not better. D is contradicted by the demo itself.

Q19 [Type: FRQ -Short Answer | Difficulty: Hard | QID : 54049]

Scenario: Below are two systems used in everyday life:

System A -A weather app that follows the rule: "If humidity > 85% AND pressure is dropping, predict rain."

System B -Spotify's Discover Weekly playlist, which is generated based on the songs you have listened to, skipped, and saved.

Question: Identify which system is rule-based and which is learning-based, and explain in 2–3 sentences how you can tell.

(Respond in 3–4 sentences.)

Model Answer: System A is rule-based: the humidity and pressure thresholds were written by humans, and the app does exactly the same thing every time regardless of who is using it. System B is learning-based: Spotify's recommendations are different for every user because the model is finding patterns in each person's listening behaviour and comparing it to similar users. The clearest test is whether the system improves over time with new examples (ML) or does the same thing every time (rule-based) -and whether a human wrote the conditions explicitly or the system discovered them from data.

Q20 [Type: FRQ -Scenario Analysis | Difficulty: Hard | QID : 54050]

Scenario: A medical team is building a Machine Learning model to detect skin conditions from photographs. They have access to 10,000 high-quality photos, but every photo in the dataset is of light skin. The team is excited about deploying the model in clinics around the world.

Question: Identify the most significant problem with this setup, explain why it matters using ideas from this session (such as patterns, generalization, or training data variety), AND describe one concrete step the team should take before deploying the model widely.

(Respond in 3–4 sentences.)

Model Answer: The most significant problem is that the training data lacks variety in skin tone, so the model will likely fail on patients with darker skin even though it appears accurate during testing. This matters because ML models can only recognise patterns similar to what they saw during training -and a model that learns "skin condition" from one skin tone alone will not generalise to skin it has never been trained on. The session's repeated warning about variety in training data applies directly here: if every example looks the same in some way, the model latches onto that sameness as part of the pattern. Before deploying widely, the team should collect a much more diverse training set that includes photos across the full range of skin tones, retrain the model, and then specifically test its accuracy broken down by skin tone before trusting it in clinics.